PROJECT TITLE: SWIFTSHIP TRACKER

COLLEGE NAME: ARJ COLLEGE OF ENGINEERING AND TECHNOLOGY

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TEAM LEADER: RITHIKA V

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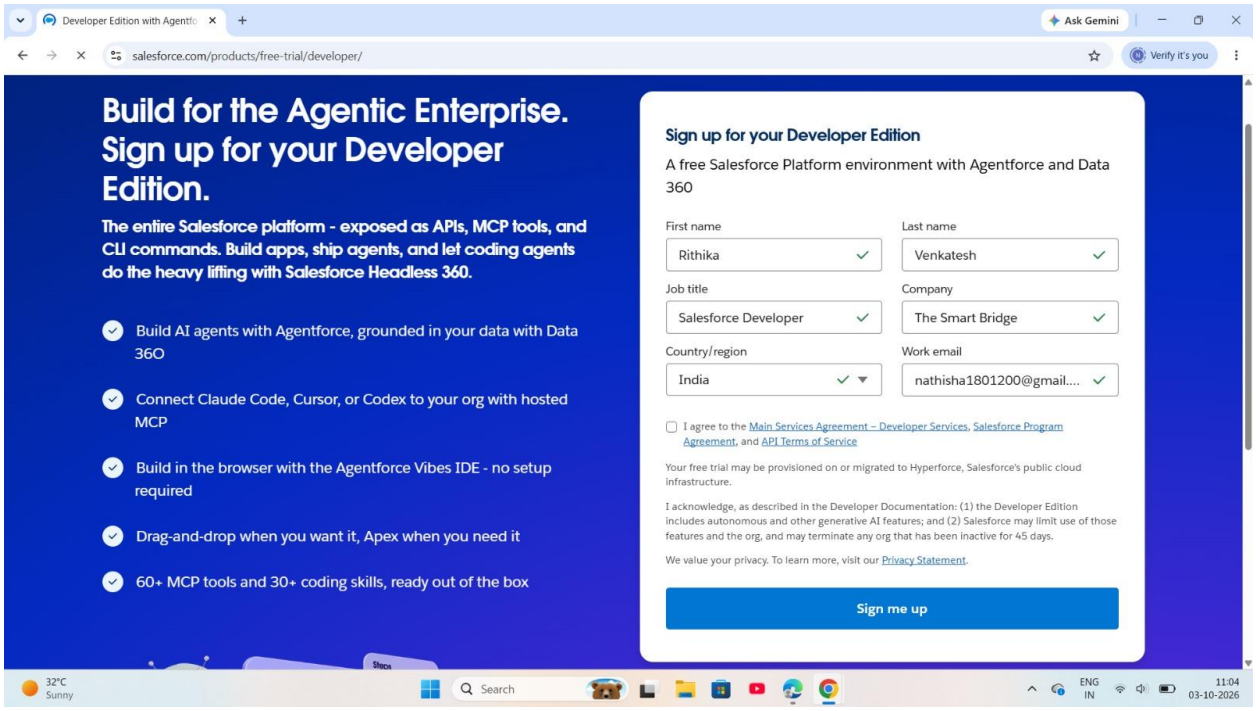
TEAM MEMBERS2: DHARANI DURAI

TEAM MEMBER3: MAARAN

TEAM MEMBER4: THIYANESWAREN

SWIFTSHIP TRACKER – SALESFORCE BASED DELIVERY TRACKING SYSTEM

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering.



Technologies Used: Salesforce, Apex, Lightning Web Components, SOQL, Salesforce Automation.

DECLARATION

We hereby declare that the project entitled “SwiftShip Tracker – Salesforce Based Delivery Tracking System” is our original work carried out as part of our academic project. The project focuses on developing a Salesforce-based system for managing shipment details, tracking delivery status and improving delivery management.

The information presented in this report is based on the implementation and testing performed during the project.

ACKNOWLEDGEMENT

We express our sincere gratitude to our institution, department and faculty members for providing us with the necessary facilities and guidance to complete the SwiftShip Tracker project.

We thank our project guide for valuable suggestions, continuous support and encouragement throughout the development of this project.

We also thank our friends and teammates for their cooperation and support during the implementation and testing stages.

ABSTRACT

SwiftShip Tracker is a Salesforce-based delivery tracking and management system designed to simplify shipment management and tracking activities.

The system provides a centralized platform for maintaining shipment information, customer details, delivery status and related records. Salesforce features such as custom objects, fields, relationships, permissions, automation, Apex and Lightning Web Components are used to develop the application.

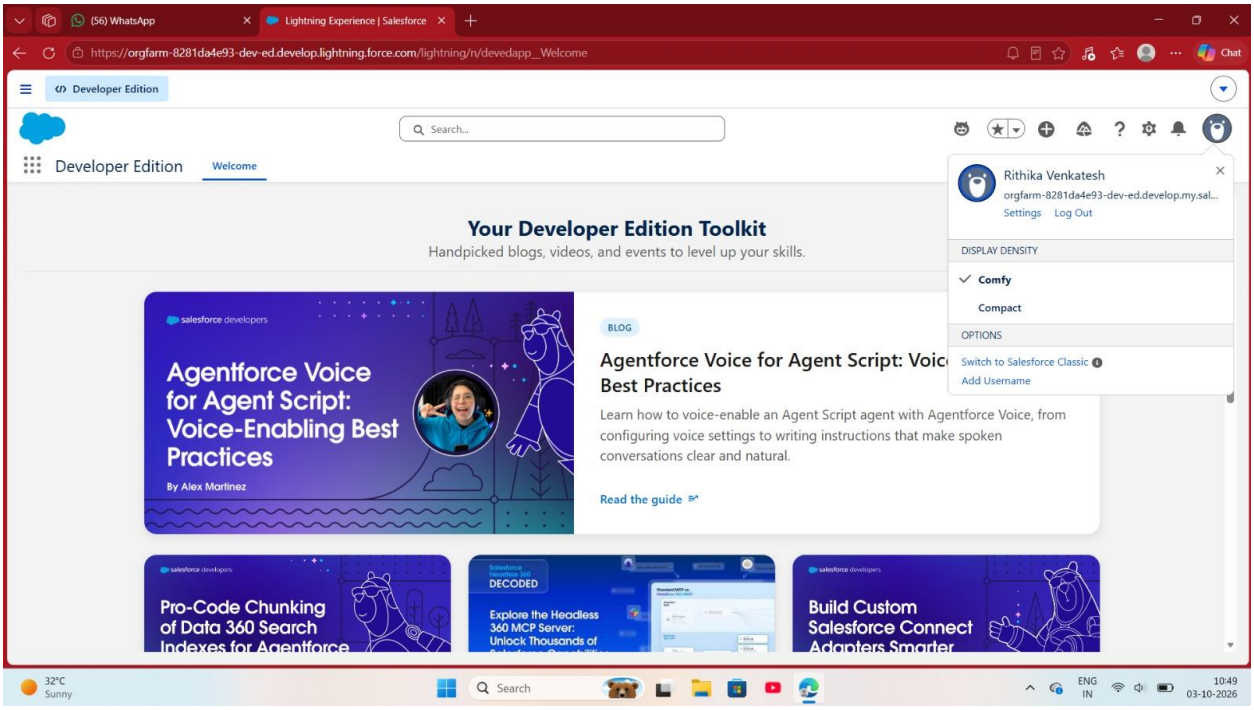
The main purpose of the system is to reduce manual work, improve data accessibility and provide a structured method for monitoring shipments from creation to delivery.

INTRODUCTION

In traditional delivery management systems, shipment information may be maintained using multiple applications or manual records. This can make tracking difficult and may result in delays or inaccurate information.

SwiftShip Tracker uses the Salesforce platform to provide a centralized solution for shipment management. Users can create and maintain shipment records, update delivery status and access related customer and delivery information.

Salesforce provides a scalable cloud environment that allows the system to manage data securely while supporting automation and customized user interfaces.



PROBLEM STATEMENT

Managing shipments manually can involve several challenges such as:

Difficulty in maintaining shipment records.

Lack of centralized tracking information.

Manual status updates.

Difficulty in accessing customer and shipment details.

Increased possibility of data-entry errors.

Limited visibility into delivery progress.

SwiftShip Tracker addresses these problems by providing a centralized Salesforce-based shipment management system.

OBJECTIVES OF THE PROJECT

The main objectives of SwiftShip Tracker are:

1. To create a centralized shipment management system.

2. To maintain customer and shipment information.

3. To track shipment delivery status.

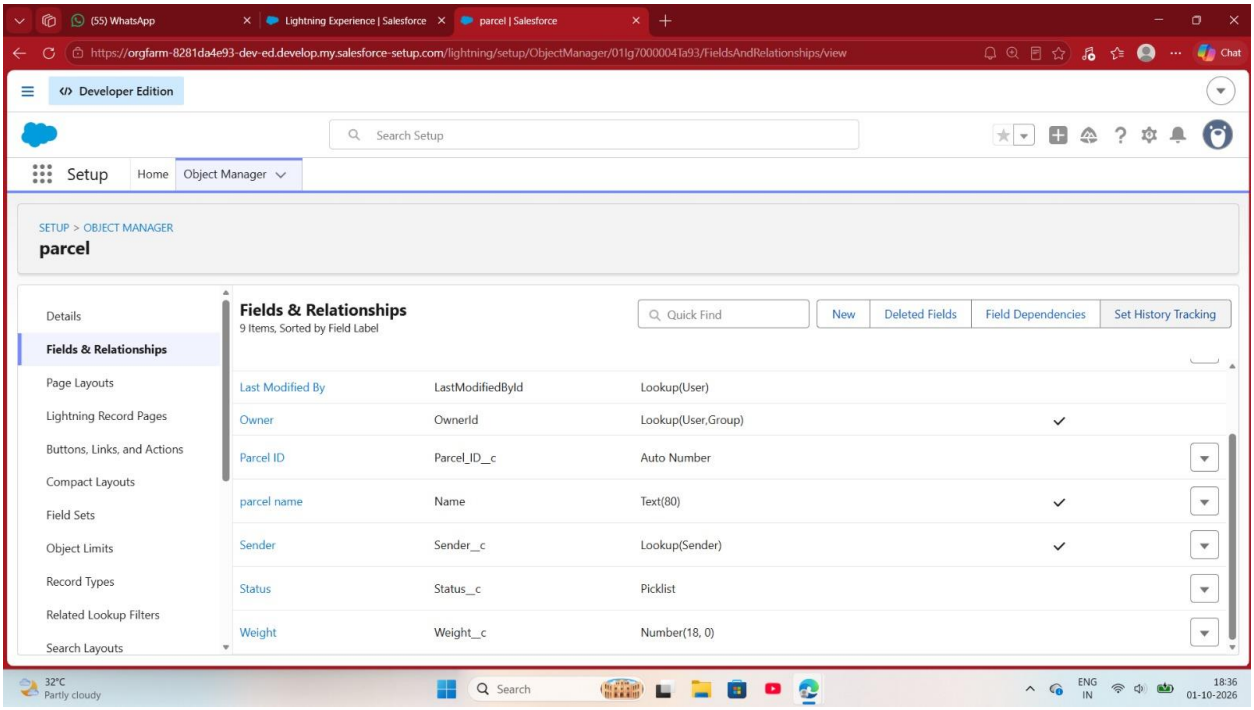
4. To establish relationships between related records.

5. To provide secure access to Salesforce users.

6. To reduce manual data management.

7. To automate repetitive processes.

8. To provide an easy-to-use interface using Lightning components.



EXISTING SYSTEM

The existing delivery management process may depend on manual records, spreadsheets or separate systems for maintaining shipment information.

Such systems can make it difficult to maintain consistent data and provide real-time visibility into shipment progress.

Common limitations include:

Manual data entry.

Difficult record searching.

Data duplication.

Limited automation.

Difficult status monitoring.

Less centralized information.

PROPOSED SYSTEM – SWIFTSHIP TRACKER

SwiftShip Tracker is proposed as a Salesforce-based solution for managing shipment and delivery information.

The system stores important information in Salesforce objects and allows users to manage records through the Salesforce interface.

The proposed system provides:

Centralized shipment records.

Customer information management.

Shipment status tracking.

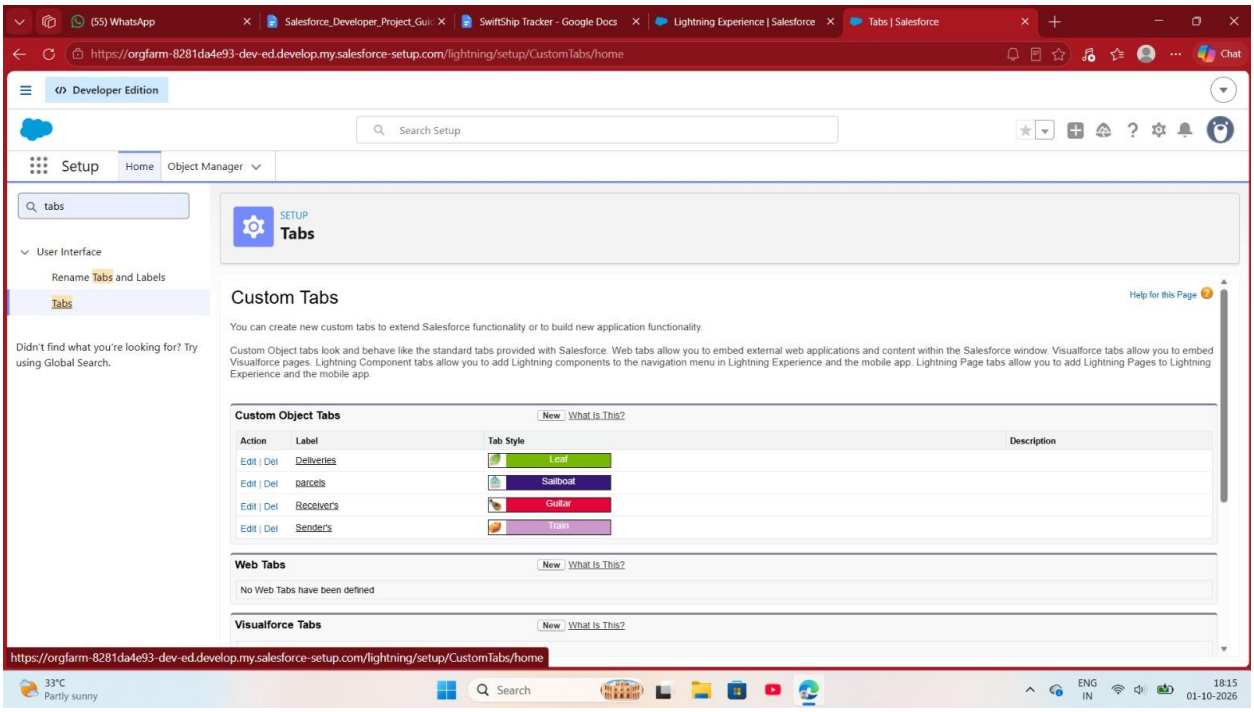
Object relationships.

Role and permission management.

Automation.

Apex-based functionality.

Lightning Web Component interface.



SALESFORCE ARCHITECTURE

Salesforce follows a cloud-based architecture where application data and business logic are managed through the Salesforce platform.

The SwiftShip Tracker project uses different Salesforce layers including:

Data Layer – Objects, fields and relationships.

Business Logic Layer – Apex and automation.

Security Layer – Profiles, permissions and sharing.

Presentation Layer – Lightning Experience and Lightning Web Components.

This architecture helps maintain the application in a structured and scalable manner.

OBJECTS AND FIELDS

Custom objects and fields are used to store SwiftShip Tracker information.

The objects represent different entities involved in shipment management. Fields store individual information such as shipment details, customer information, delivery status and other required attributes.

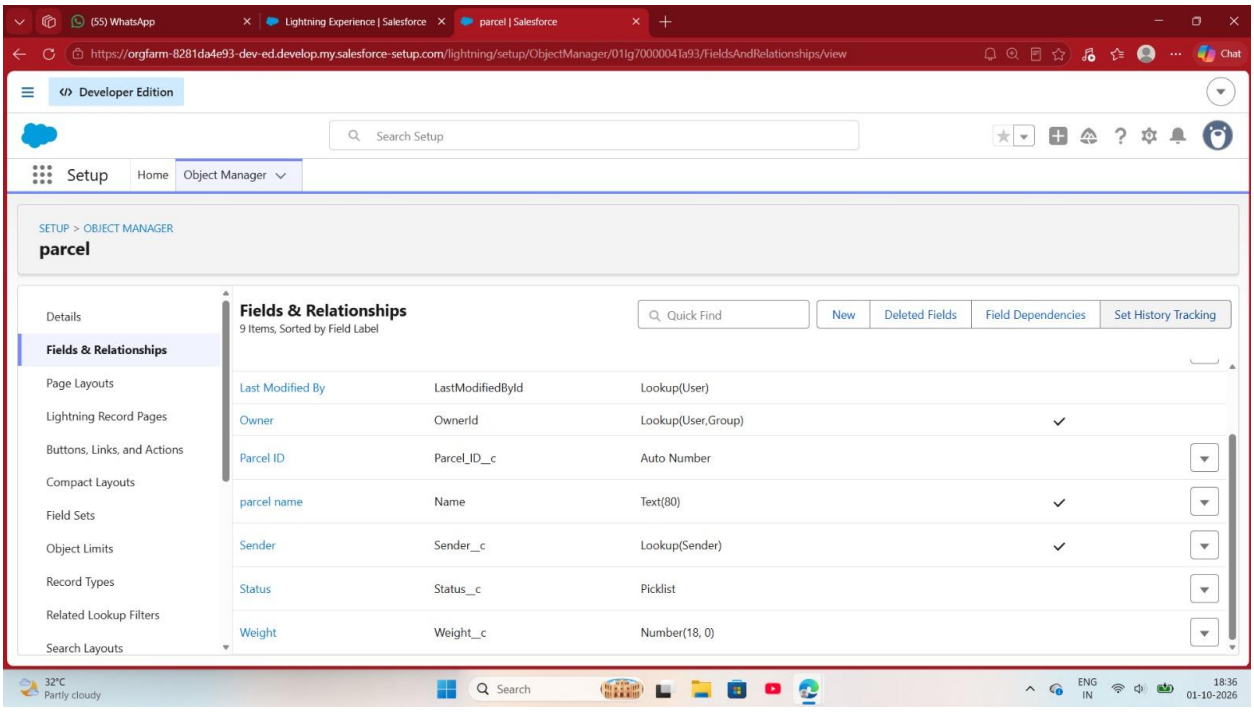
Relationships between objects allow related information to be accessed together.

DATA MODEL

The data model defines how different records in the SwiftShip Tracker application are connected.

Relationships are used to associate shipment records with relevant customer and delivery information.

Using Salesforce relationships improves data organization and allows users to navigate between related records easily.



SECURITY AND ACCESS CONTROL

Salesforce security features are used to control access to SwiftShip Tracker data.

The project uses Salesforce security mechanisms such as:

Profiles

Permission Sets

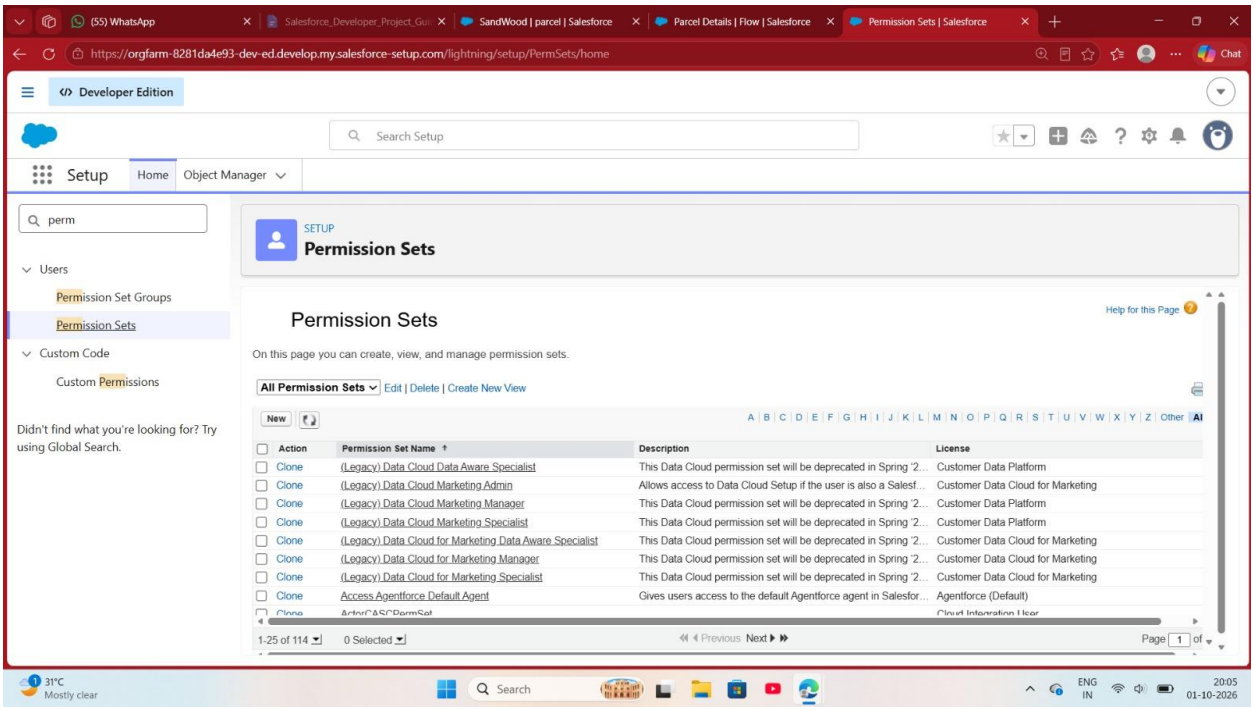
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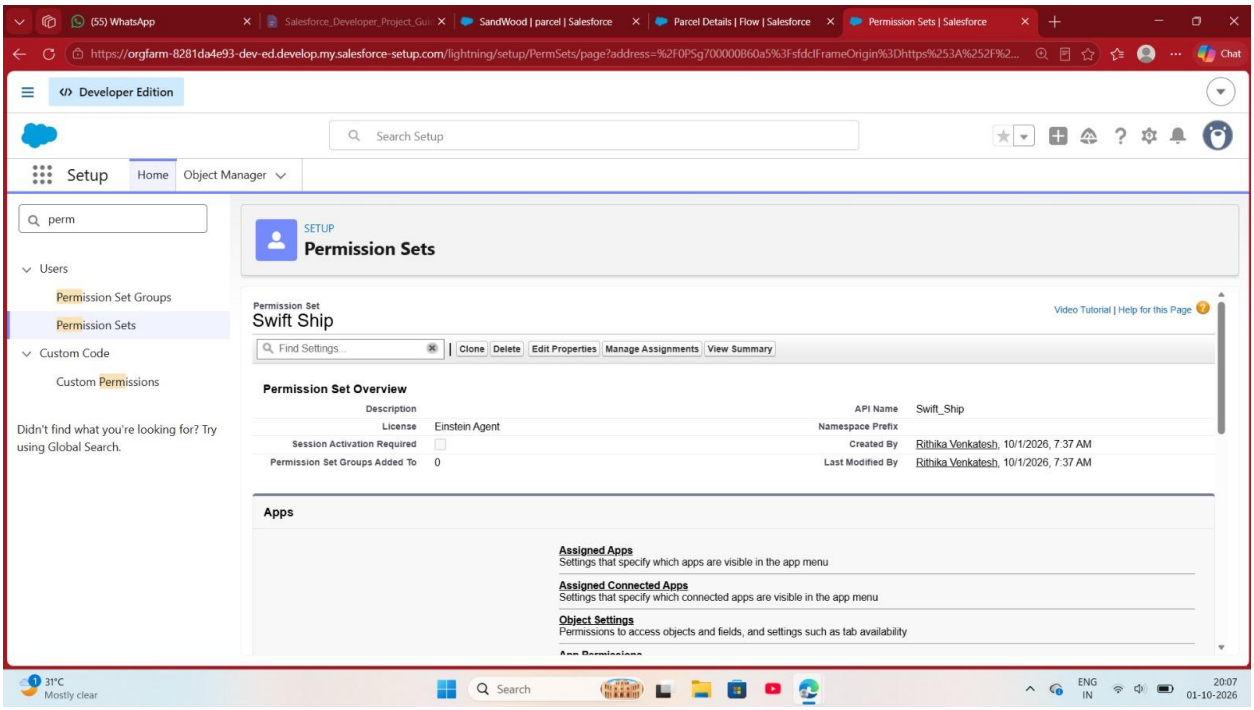
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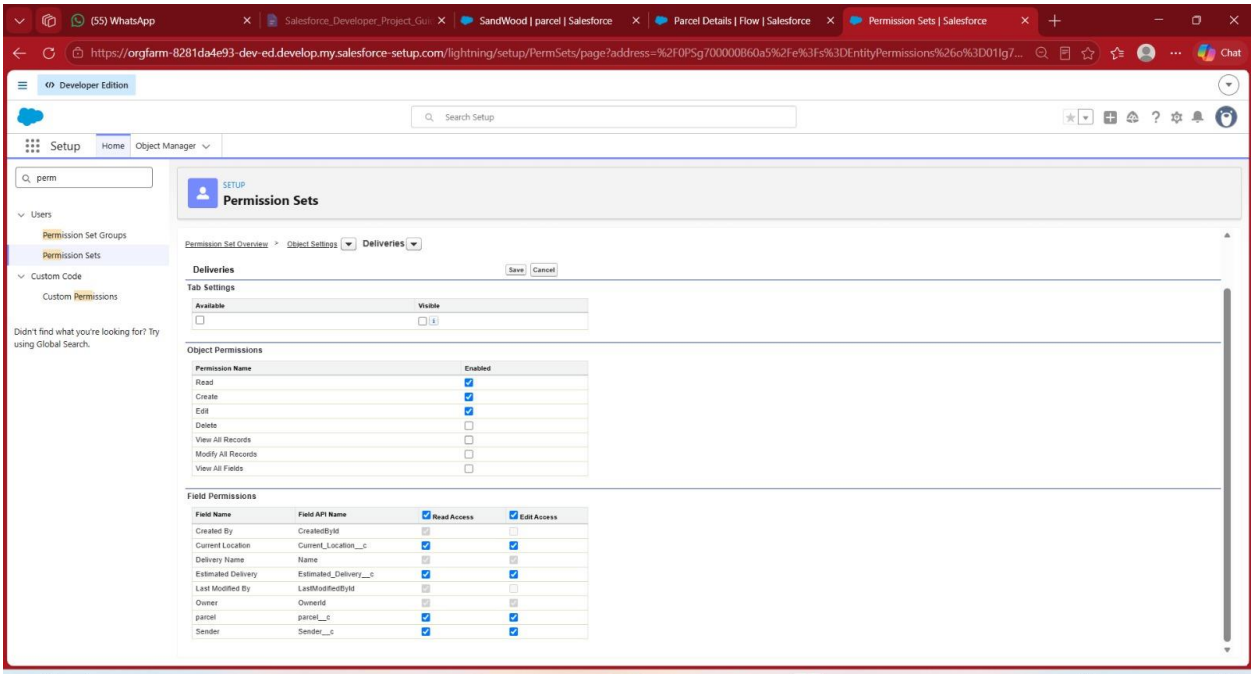
Field-Level Security

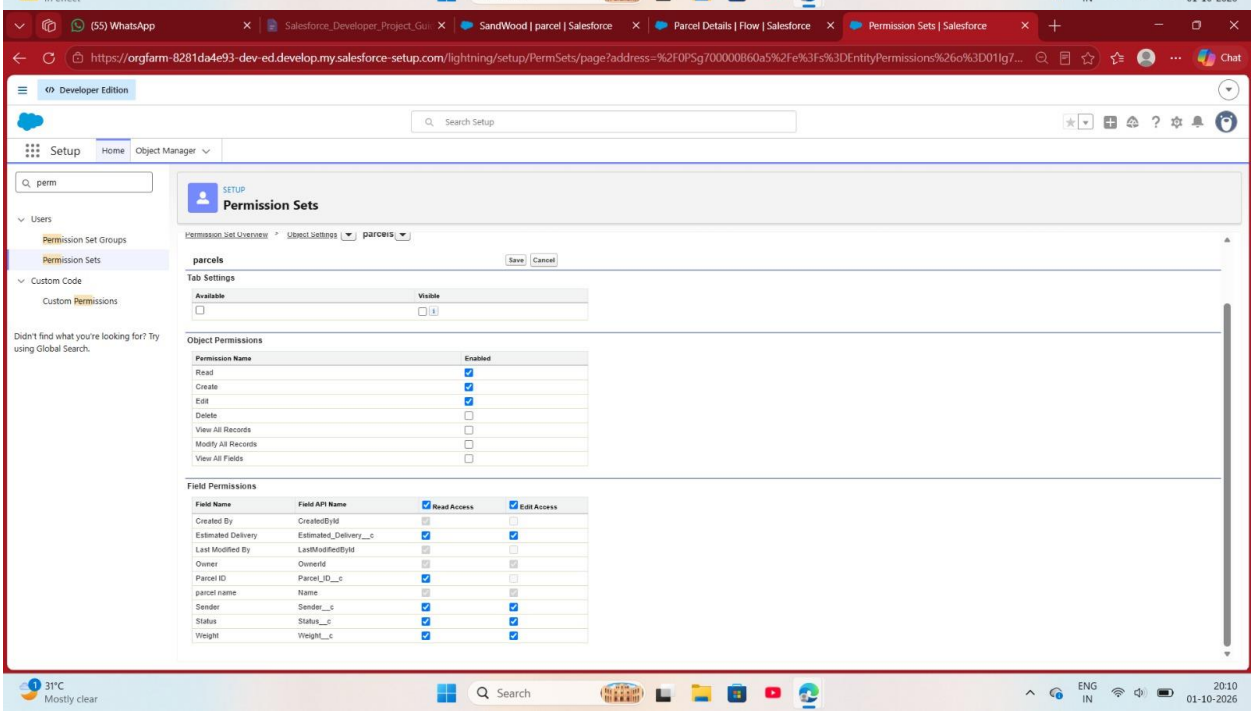
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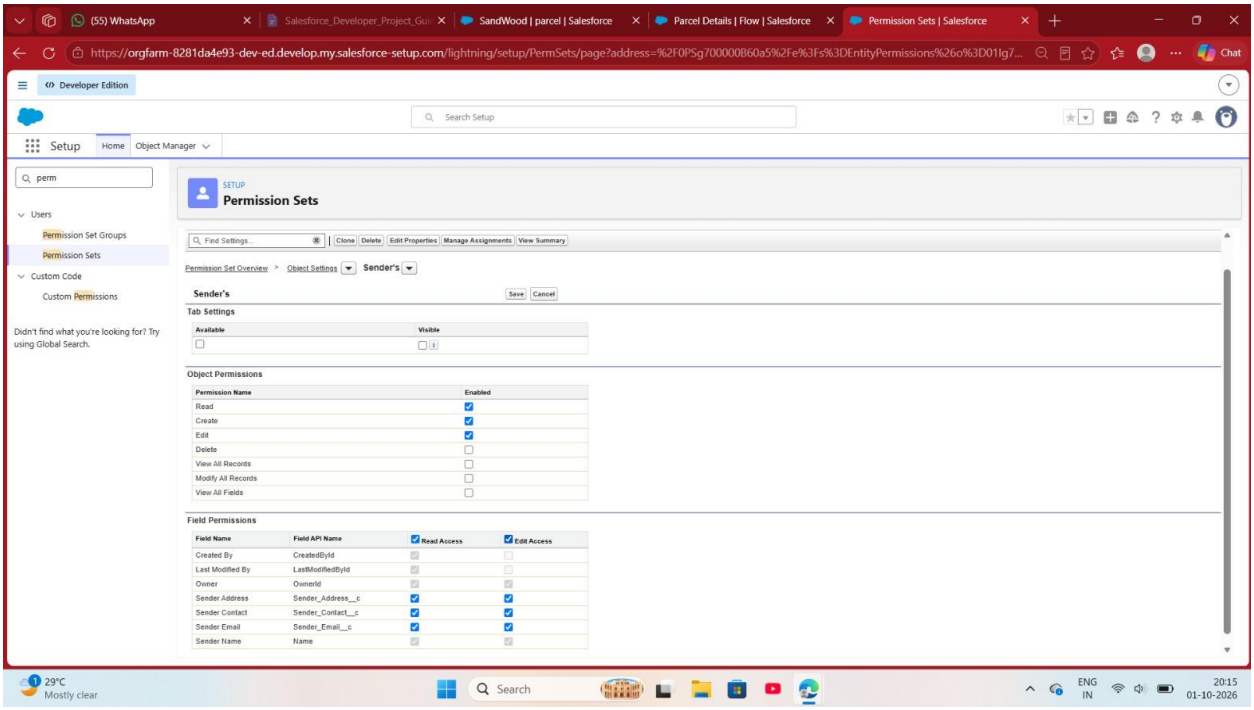
These features ensure that users can access only the information and operations required for their responsibilities.

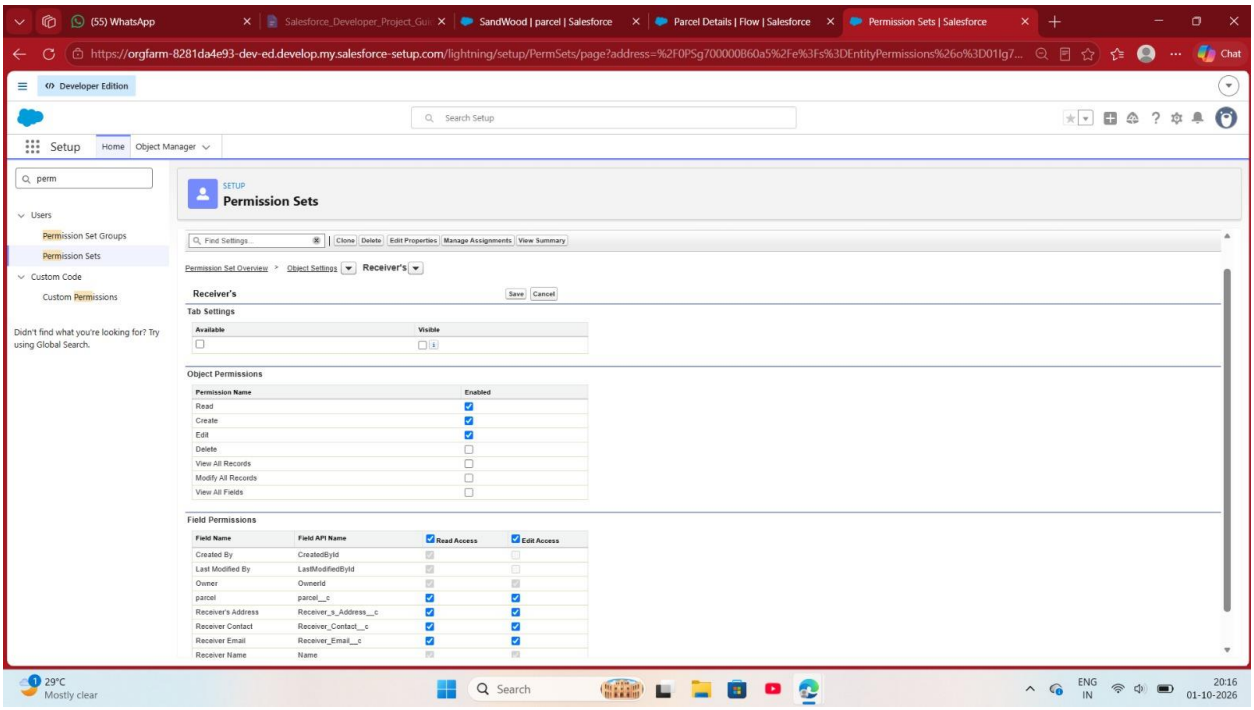

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APEX AND AUTOMATION

Apex is Salesforce's programming language used to implement server-side business logic.

In SwiftShip Tracker, Apex can be used for processing shipment-related operations and implementing customized functionality that cannot be achieved using standard Salesforce configuration alone.

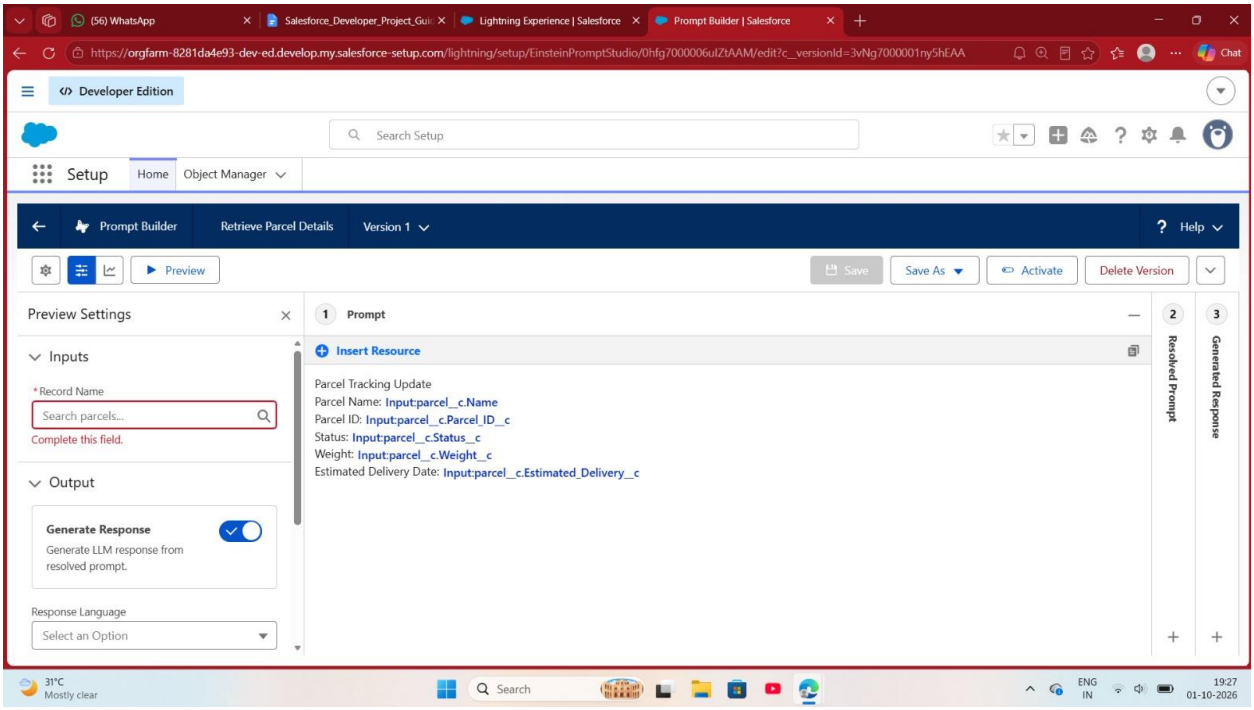
Automation helps reduce repetitive manual activities and improves consistency in record processing.

LIGHTNING WEB COMPONENTS

Lightning Web Components (LWC) are used to create customized and interactive user interfaces in Salesforce.

The SwiftShip Tracker project uses LWC to display application information in a user-friendly manner.

LWC provides reusable components that can interact with Salesforce data and provide users with a modern interface.



SYSTEM IMPLEMENTATION

The SwiftShip Tracker system was implemented using Salesforce development and configuration features.

The implementation process includes:

1. Creating Salesforce objects.

2. Creating required fields.

3. Establishing relationships.

4. Configuring permissions.

5. Developing Apex functionality.

6. Creating Lightning components.

7. Configuring automation.

8. Testing the application.

Each component was tested to ensure that the system functions according to the project requirements.

TESTING AND RESULTS

Testing was performed to verify the functionality of the SwiftShip Tracker system.

The major areas tested include:

Test Area	Expected Result
Record Creation	Record should be created successfully
Field Validation	Correct values should be accepted
Shipment Status	Status should update correctly
Permissions	Users should access permitted data
LWC	Component should display correctly
Apex	Required logic should execute successfully

The testing results demonstrate that the implemented Salesforce components work together to support shipment management.

ADVANTAGES

Centralized shipment management.

Easy access to delivery information.

Improved data organization.

Secure user access.

Reduced manual work.

Customizable Salesforce interface.

Automation of repetitive processes.

FUTURE ENHANCEMENTS

Future versions can include:

Real-time GPS tracking.

Customer notification system.

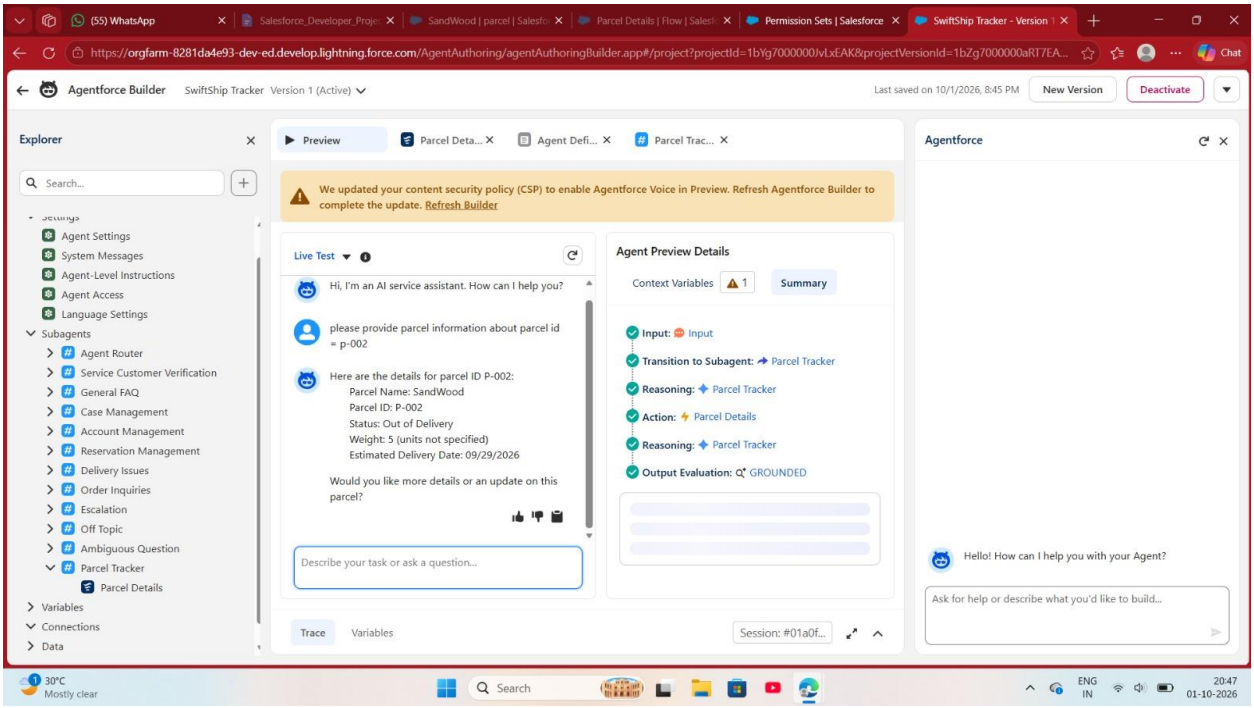
Mobile application.

Advanced delivery analytics.

AI-based delivery predictions.

Automated SMS/email notifications.

Integration with external logistics AIPs



CONCLUSION

SwiftShip Tracker demonstrates how Salesforce can be used to develop a structured delivery tracking and management application.

The project combines Salesforce configuration, data management, security, Apex, automation and Lightning Web Components to create a centralized system for shipment management.

The implementation provides a foundation that can be further enhanced with real-time tracking, notifications, analytics and external integrations.

REFERENCES

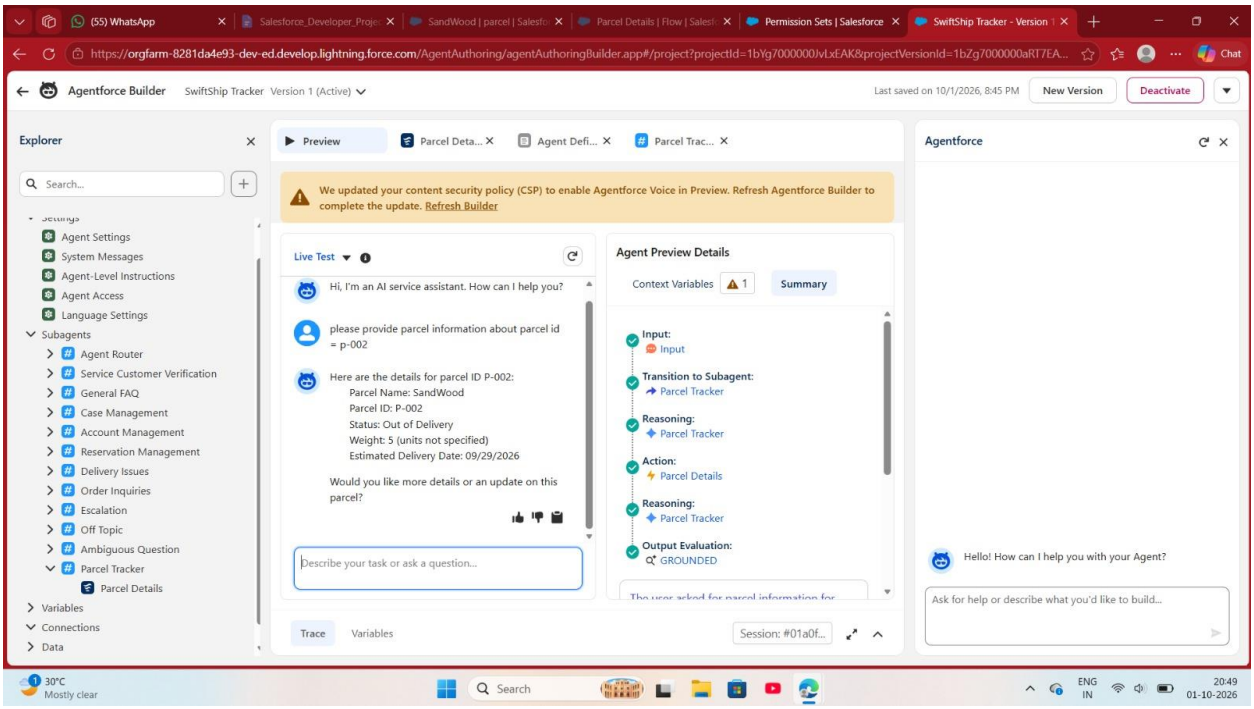
Salesforce Documentation

Salesforce Trailhead

Salesforce Developer Documentation

Apex Developer Guide

Lightning Web Components Documentation


Agentforce Builder
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Agent Settings
System Messages
Agent-Level Instructions
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Parcel Details
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We updated your content security policy (CSP) to enable Agentforce Voice in Preview. Refresh Agentforce Builder to complete the update. Refresh Builder
Live Test
Hi, I'm an AI service assistant. How can I help you?
please provide parcel information about parcel id = p-002
Here are the details for parcel ID P-002:
Parcel Name: SandWood
Parcel ID: P-002
Status: Out of Delivery
Weight: 5 (units not specified)
Estimated Delivery Date: 09/29/2026
Would you like more details or an update on this parcel?
Describe your task or ask a question...
Agent Preview Details
Context Variables
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Summary
Input:
Input
Transition to Subagent:
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